QUBO Formulation: Minimum Dominating Set Problem

Minimum Dominating Set Problem

Problem Statement. Given an undirected graph $G = (V, E)$, the Minimum Dominating Set problem seeks a subset of vertices $D \subseteq V$ of minimum cardinality such that every vertex in V is either contained in D or adjacent to at least one vertex in D. Formally, we require that for every vertex $v \in V$, there exists a vertex $u \in D$ such that $u = v$ or $(u, v) \in E$.

Decision Variables. We introduce a binary variable $x_i \in \{0, 1\}$ for each vertex $i \in V$. The variable x_i indicates membership in the dominating set:

$$x_i = \begin{cases} 1 & \text{if vertex } i \in D, \\ 0 & \text{otherwise.} \end{cases}$$

QUBO Derivation. We construct a Quadratic Unconstrained Binary Optimization (QUBO) formulation by encoding the objective and constraints into a single Hamiltonian $H(\mathbf{x})$.

Step 1: Objective Function. The goal is to minimize the size of the dominating set, which corresponds to minimizing the sum of the decision variables:

$$\min \sum_{i \in V} x_i. \tag{1}$$

Step 2: Constraints. For each vertex $j \in V$, let $N(j) = \{i \in V \mid (i, j) \in E\}$ denote the set of neighbors of j. The domination requirement is that j must be selected or adjacent to a selected vertex:

$$x_j + \sum_{i \in N(j)} x_i \geq 1, \quad \forall j \in V. \tag{2}$$

Define the local domination sum S_j as:

$$S_j = x_j + \sum_{i \in N(j)} x_i. \tag{3}$$

The constraint is satisfied if and only if $S_j \geq 1$. Since x_i are binary, S_j is a non-negative integer. A violation occurs only when $S_j = 0$.

Step 3: Penalty Terms. We enforce the constraints using quadratic penalty terms with weight $\lambda > 0$. For each $j \in V$, we add:

$$P_j(\mathbf{x}) = \lambda \left(1 - S_j\right)^2. \tag{4}$$

This term is zero when $S_j \geq 1$ and equals λ when $S_j = 0$. Expanding P_j:

$$P_j(\mathbf{x}) = \lambda \left(1 - 2S_j + S_j^2\right). \tag{5}$$

Using the idempotent property $x_k^2 = x_k$ for binary variables, we expand S_j^2:

$$S_j^2 = \left(x_j + \sum_{i \in N(j)} x_i\right)^2 \tag{6}$$

$$= x_j^2 + 2x_j \sum_{i \in N(j)} x_i + \left(\sum_{i \in N(j)} x_i\right)^2 \tag{7}$$

$$= x_j + 2 \sum_{i \in N(j)} x_j x_i + \sum_{i \in N(j)} x_i^2 + \sum_{\substack{i,k \in N(j) \\ i \neq k}} x_i x_k \tag{8}$$

$$= x_j + 2 \sum_{i \in N(j)} x_j x_i + \sum_{i \in N(j)} x_i + \sum_{\substack{i,k \in N(j) \\ i \neq k}} x_i x_k. \tag{9}$$

Substituting this back into the penalty expression:

$$P_j(\mathbf{x}) = \lambda \left[1 - 2\left(x_j + \sum_{i \in N(j)} x_i\right) + x_j + 2 \sum_{i \in N(j)} x_j x_i + \sum_{i \in N(j)} x_i + \sum_{\substack{i,k \in N(j) \\ i \neq k}} x_i x_k\right] \tag{10}$$

$$= \lambda \left[1 - x_j - \sum_{i \in N(j)} x_i + 2 \sum_{i \in N(j)} x_j x_i + \sum_{\substack{i,k \in N(j) \\ i \neq k}} x_i x_k\right]. \tag{11}$$

Step 4: Combined Hamiltonian. The total QUBO Hamiltonian is the sum of the objective and all penalty terms:

$$H(\mathbf{x}) = \sum_{i \in V} x_i + \sum_{j \in V} P_j(\mathbf{x}). \tag{12}$$

Collecting terms, we obtain:

$$H(\mathbf{x}) = c + \sum_{i \in V} Q_{i,i} x_i + \sum_{\substack{i,j \in V \\ i < j}} Q_{i,j} x_i x_j, \tag{13}$$

where the constant offset c and coefficients $Q_{i,j}$ are derived below.

Step 5: QUBO Coefficients. Summing the linear contributions from P_j over all $j \in V$:

$$\sum_{j \in V} \left(-\lambda x_j - \lambda \sum_{i \in N(j)} x_i \right) = -\lambda \sum_{j \in V} x_j - \lambda \sum_{j \in V} \sum_{i \in N(j)} x_i \tag{14}$$

$$= -\lambda \sum_{i \in V} x_i - \lambda \sum_{i \in V} x_i \deg(i) \tag{15}$$

$$= -\lambda \sum_{i \in V} (1 + \deg(i))\, x_i, \tag{16}$$

where $\deg(i) = |N(i)|$ is the degree of vertex i. Adding the objective term $\sum_{i \in V} x_i$, the diagonal QUBO entries are:

$$Q_{i,i} = 1 - \lambda\,(1 + \deg(i))\,, \quad \forall i \in V. \tag{17}$$

For off-diagonal entries $i \neq j$, contributions arise from the quadratic terms in P_j. The term $x_i x_j$ appears in P_k if $i, j \in N(k)$ or if $j \in N(i)$ or $i \in N(j)$. Specifically:

- If $(i, j) \in E$, the term $2\lambda x_i x_j$ appears in P_i and P_j, contributing 4λ.
- If $(i, j) \notin E$, the term $x_i x_j$ appears in P_k for each common neighbor $k \in N(i) \cap N(j)$. Let $C_{ij} = |N(i) \cap N(j)|$ be the number of common neighbors. Each such k contributes $2\lambda x_i x_j$ via the cross-term in S_k^2.

Thus, for $i < j$:

$$Q_{i,j} = \begin{cases} 4\lambda & \text{if } (i, j) \in E, \\ 2\lambda\, C_{ij} & \text{if } (i, j) \notin E. \end{cases} \tag{18}$$

The constant offset is the sum of constants from all penalties:

$$c = \sum_{j \in V} \lambda = \lambda |V|. \tag{19}$$

Penalty Weight Justification. The penalty weight λ must be chosen large enough to ensure that any infeasible solution has a strictly higher energy than the optimal feasible solution. The maximum possible value of the objective function is $|V|$. Violating a single constraint increases the penalty by at least λ, while the maximum reduction in the objective from violating constraints is bounded by $|V|$. Therefore, choosing $\lambda > |V|$ guarantees that the penalty for any infeasible assignment dominates any potential gain in the objective, ensuring the global minimum of $H(\mathbf{x})$ corresponds to a feasible and optimal dominating set.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\mathbf{x} \in \{0, 1\}^{|V|}$ recovers the optimal dominating set because:

1. *Feasibility:* For any feasible solution, all constraints are satisfied, so all penalty terms vanish. The energy equals the objective value.

2. *Optimality:* For any infeasible solution, at least one constraint is violated, incurring a penalty of at least λ. Since $\lambda > |V| \geq$ objective value for any solution, the energy of any infeasible solution exceeds the energy of any feasible solution. Thus, the global minimum must be feasible. Among feasible solutions, $H(\mathbf{x})$ reduces to the objective function, so the minimum corresponds to the minimum dominating set.

Illustrative Example. Consider a path graph with $N = 3$ vertices $V = \{0, 1, 2\}$ and edges $E = \{(0, 1), (1, 2)\}$. The degrees are $\deg(0) = 1$, $\deg(1) = 2$, $\deg(2) = 1$. The number of common neighbors is $C_{02} = 1$ (vertex 1), and $C_{01} = C_{12} = 0$. We choose $\lambda = 10$, satisfying $\lambda > N$.

Using the general formulas:

$$Q_{0,0} = 1 - 10(1 + 1) = -19, \tag{20}$$
$$Q_{1,1} = 1 - 10(1 + 2) = -29, \tag{21}$$
$$Q_{2,2} = 1 - 10(1 + 1) = -19, \tag{22}$$
$$Q_{0,1} = 4 \times 10 = 40, \tag{23}$$
$$Q_{1,2} = 4 \times 10 = 40, \tag{24}$$
$$Q_{0,2} = 2 \times 10 \times 1 = 20, \tag{25}$$
$$c = 10 \times 3 = 30. \tag{26}$$

The resulting QUBO Hamiltonian is:

$$H(x_0, x_1, x_2) = 30 - 19x_0 - 29x_1 - 19x_2 + 40x_0x_1 + 40x_1x_2 + 20x_0x_2. \tag{27}$$

Evaluating H for candidate solutions:

- $\mathbf{x} = (0, 1, 0)$: $H = 30 - 29 = 1$. This corresponds to dominating set $\{1\}$, size 1.
- $\mathbf{x} = (1, 0, 1)$: $H = 30 - 19 - 19 + 20 = 12$. This corresponds to dominating set $\{0, 2\}$, size 2.
- $\mathbf{x} = (0, 0, 0)$: $H = 30$. Infeasible.

The minimum energy is 1, achieved at $\mathbf{x} = (0, 1, 0)$, correctly identifying the minimum dominating set $\{1\}$.